Feasibility Analysis & MVP Roadmap

Project: Climate Response Hub

Team: Program Counter

1. Executive Summary

Climate Response Hub is a decision-support dashboard designed to assist emergency management coordinators during extreme heatwaves and outage-related climate events. By integrating fragmented data - including climate risk, vulnerability indicators, and cooling center availability - into a unified operational view, the platform reduces cognitive load and accelerates crisis triage. It acts as a planning and coordination layer that does not directly control infrastructure, ensuring a high degree of operational feasibility for municipal adoption.

2. Project Vision & Problem Statement

- **Problem**: Climate-related emergencies are increasing, yet information is fragmented across multiple sources, slowing down critical decision-making.
- **Goal**: To help coordinators identify high-risk neighborhoods, prioritize support, and coordinate response actions more effectively.
- **Impact**: Enhances community resilience and equity by ensuring limited emergency resources are targeted where they are most needed.

3. Solution Overview

The dashboard provides an interactive interface for coordinators to:

- Visualize neighborhood-level climate and outage vulnerability on an interactive map.
- Analyze risk using combined weather data and vulnerability indicators.
- Log, track, and update response actions in real-time, preventing duplicate efforts.

Core Technical Architecture

The platform is built as a lightweight, frontend-focused MVP:

Layer	Technology
Frontend	React
GIS / Mapping	ArcGIS / Esri
Data Format	GeoJSON, JSON
Data Storage	localStorage (MVP) / Potential for MongoDB (Future)

4. Feasibility Analysis

Operational

The tool fits into existing workflows without replacing human decision-makers, keeping the scope realistic for emergency coordinators.

Economic

Uses open data (City of Toronto Open Data) and common web technologies, requiring very low initial capital for deployment.

Technical

Employs rule-based triage logic and spatial analysis (via ArcGIS API) to identify localized hotspots. The MVP uses Mock APIs for infrastructure capacity (e.g., cooling center load), which are designed for seamless transition to live utility APIs in production.

5. MVP Scope & Strategic Roadmap

MVP Includes

- Interactive neighborhood map and risk visualization.
- Rule-based triage logic and detail panels for neighborhood insights.
- Prototype response action logging.

6-Month Roadmap

- Phase 1 (Months 0-1): Finalize MVP, GIS integration, and local CRUD workflow (Current Stage).
- Phase 2 (Months 2-3): Transition to robust database (MongoDB) with RBAC and integration of live utility APIs (e.g., Alectra Utilities).

- Phase 3 (Months 4-6): Expand to other climate risks (e.g., winter cold, floods) and develop Energy Equity reporting modules.

6. Strategic Impact

This project aligns with the Government of Canada's 2026 and 2040 climate adaptation goals. By providing visibility into backup power and community vulnerability, the platform supports corporate resilience targets for partners like Alectra Utilities.

Note: This platform does not provide real-time power grid control, direct dispatch of emergency services, or real-time citizen-facing alerts. Its purpose is to serve as a coordination hub to enhance human-led planning.